Does Money Move Votes in Washington?

A negative result, and the data ceiling that keeps it from being a stronger one

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AI-assisted drafting and analysis review. All figures are reproducible from public-record data and from the open-source scripts cited below. The paper source, code, and data-acquisition recipe are public at https://github.com/skirby359/who-decides. *Contact: kirby@tikorconsulting.com.*

*Paper #4 of the electoral-health series (companion to `who-decides-washington.md`, `safe-seat-washington.md`, `donor-class-and-the-electorate.md`, and `cross-state-fec-money.md`). DRAFT — pending the independent-verification gate in `publication-checklist.md`.***

Abstract

The three preceding papers in this series establish who votes, whether their vote decides anything, and who funds the candidates. This paper asks the question those invite — does the money change the result? — and reports that the available public record cannot answer it, while characterizing precisely how far short it falls. Across 163 Washington legislative and congressional races from 2018 to 2024, the ratio of Democratic to Republican fundraising correlates with candidate overperformance at **+0.55**, the strongest of any measured factor, ahead of incumbency (+0.43) and candidate quality (+0.34). Three independent attempts to convert that correlation into a causal reading fail. Spending *allocation* — the field, media, and professional shares of itemized expenditure — has cross-cycle holdout R^2 of essentially **zero**, despite being a stable candidate trait. The forecast model that underpins this series zeroes its fundraising term in post-redistricting districts because the single-cycle baseline already absorbs it. And the one design capable of a directional test, regressing fundamentals-net residual on net independent-expenditure advantage, is not estimable: machine-readable directional IE exists for a single cycle and seven scorable Washington races, while $70.6M of state legislative IE carries no support-or-oppose flag at all. The descriptive cross-section that remains points the wrong way for a purchase story — money is associated with running *behind* — and the most IE-saturated House race in the country finished 0.06 points from its fundamentals. The honest verdict is that money in Washington behaves as a marker of candidate strength rather than a demonstrable mover of votes, and that the public data cannot presently distinguish the two.

Keywords: campaign spending; independent expenditures; electoral effects; endogeneity; null results; data availability; state legislative elections; Washington.

The question, and why it resists an answer

The preceding papers in this series each end at the same edge. The turnout paper shows an older, smaller electorate decides off-cycle races. The safe-seat paper shows most general elections are settled before November. The donor paper shows the people funding those races are not the people voting in them. Each invites the obvious next question: **so does the money actually change who wins?**

It is the hardest question in the series, and the reason is structural rather than technical. Campaign money is *endogenous to candidate strength*. Strong candidates attract money; money then appears alongside strong results. Donors and independent-expenditure committees also behave strategically, moving resources toward races they judge close and toward candidates they judge threatened — which can invert the naive correlation entirely, making spending look harmful. Untangling this is a fifty-year-old problem in political science (Jacobson 1978), and the credible answers come from designs this data does not support: instrumental variables (Gerber 1998), repeat-challenger panels (Levitt 1994), or randomized field experiments (Kalla & Broockman 2018).

This paper does not solve it. What it does is report what Washington's public record shows, run the tests that record permits, and state precisely which test would settle the matter and why it cannot currently be run. **The negative result and the data ceiling are the contribution.** A reader looking for a verdict on whether money buys elections will not find one here, and should be suspicious of anyone offering one from this evidence base.

Finding 1 — Money is the single strongest correlate of overperformance

The starting observation is not subtle. Across the 163 baseline-scorable Washington legislative and congressional races from

2018 to 2024 (`scripts/diag_overperformance_patterns.py`), *overperformance* — the actual Democratic two-party share minus the district's fundamentals-based baseline — correlates with the fundraising ratio more strongly than with anything else measured:

factor	Pearson r with overperformance
fundraising, log2(D receipts / R receipts)	**+0.55**
incumbency	+0.43
candidate quality index	+0.34
local trend	+0.32
midterm year	≈0

The relationship is monotonic, not driven by a tail. Sorting races by who out-raised whom: where the Democrat out-raised the Republican, the Democrat beat the district baseline by an average of **+4.20** points; where funding was even, **+2.37**; where the Republican out-raised, **−1.93**.

Taken alone this looks like a straightforward answer, and it is the number a campaign consultant would quote. The rest of this paper is about why it is not one.

Finding 2 — Three tests for a causal signature, three nulls

If money moves votes, the effect should leave traces beyond a raw correlation. Three independent tests look for those traces. None finds one.

2a. How money is spent predicts nothing

If spending buys votes, *how* it is spent should matter — a campaign that puts its money into field organizing should perform differently from one that puts it into television. Washington's Public Disclosure Commission publishes itemized expenditures with a purpose code per transaction (Socrata `tijg-9zyp`; 272,320 candidate rows 2018–2024, $286M, ~62% of dollars carrying a code). From these, `scripts/diag_expenditures_vs_residual.py` derives each candidate's field, media, and professional *shares* of coded operational spend, and tests them against the residual.

The shares correlate with the residual at essentially zero — field **+0.02**, media **+0.05**, professional **−0.03**. Total spend correlates at +0.26, but that is the fundraising scale signal from Finding 1 arriving again, not allocation. The decisive test is cross-cycle holdout: fit on 2022, predict 2024.

model	holdout R^2
core candidate-quality features	0.000
core + field share	0.006
core + all allocation shares	0.018
allocation shares alone	0.041 *(r = −0.20, wrong-signed)*

In-sample R^2 rises from 0.049 to 0.144 as shares are added while holdout R^2 stays at the floor — the signature of overfitting, not signal. Notably, allocation *is* a real and stable candidate trait: field-share persistence across cycles runs r = 0.83 and media-share r = 0.998. It is a stable feature that carries no information about the outcome. And the one directional hint runs against the folk theory: field share correlates **−0.24** with raw overperformance, so more ground spending is associated with slightly worse results, not better.

2b. The forecast model discards the money term when it can check it

A second test comes from an unusual direction: the forecasting model this series relies on was tuned without reference to this question, and it independently concluded that fundraising carries no information once the baseline is known.

In Washington's post-2022 districts, where the redistricting boundary filter collapses history to a single post-redistricting cycle, the model **zeroes its fundraising-advantage term entirely**. The reason is that the single-cycle baseline already absorbs it: the incumbent ran in that cycle, their fundraising advantage manifested in that result, and adding the term on top double-counts. Before this correction the model predicted a D+28 result in a district whose incumbent's actual history ran D+5 to D+8. The term was not dropped for ideological reasons but because leaving it in produced worse forecasts.

This is weaker evidence than a designed test — it is a modelling decision, not an experiment — but it points the same way. When a predictive system is allowed to check whether fundraising adds information beyond the district's own recent result, it finds that it does not.

2c. Money flows toward the side that is behind

The third test is the closest thing to a directional design the data permits. `scripts/diag_ie_vs_margin.py` regresses each race's *fundamentals-net residual* — actual minus model-predicted Democratic share, not the raw margin — on the net pro-Democratic independent-expenditure advantage. Using the residual rather than the margin is what makes this a test of money's marginal effect rather than a restatement of district partisanship.

The association is **negative**: −0.39 points of residual per $1M net pro-Democratic IE (Pearson r = −0.39, n = 7). Read naively this says money *costs* votes. Read correctly it is the textbook endogeneity signature: outside money flows toward the side that is struggling, and arrives in races already moving away from it.

The single most instructive case is the most heavily funded. **Washington's 3rd Congressional District in 2024 attracted $40.1M in total independent expenditure — the most IE-saturated House race in the country — with a net $16.2M advantage on the Democratic side. It finished 0.06 points from its fundamentals-based prediction.** Forty million dollars of outside money, and the race landed almost exactly where the district's underlying partisanship said it would. That is a single observation and proves nothing on its own, but it is a striking one.

Finding 3 — The test that would settle it cannot be run

The design in 2c is the right one. It is also not estimable, and that limitation is the paper's most citable result.

Directional independent-expenditure data exists on disk for exactly one cycle. FEC Schedule E carries a support-or-oppose flag and a district, which is what makes a directional test possible at all — but only the 2024 cycle is loaded, yielding **seven scorable Washington House races**. Three further districts were uncontested and drop out; 2026 has no national-environment data yet.

The state-legislative money is worse. Washington's PDC records **$70.6M** of independent expenditure in legislative races, which would multiply the sample severalfold. It carries a **null support/oppose flag**. The database records that money was spent regarding a race, not which side it was spent for. It therefore cannot enter a directional test at all without re-derivation from sponsor-level data.

Seven cross-sectional observations cannot bear the analysis the question requires. The bootstrap confidence interval, the early-versus-late spending split, and the next-cycle placebo all need either more races or more cycles. So the script **withholds inference rather than reporting a coefficient** — at n = 7 the sign flips on a single race, and a slope reported from it would be a coin flip dressed as a finding. The descriptive number above is labelled descriptive in the script's own output for that reason.

What would unlock it. Backfilling FEC Schedule E for 2018, 2020, and 2022 via `load_fec_independent_expenditures(conn, cycle=YYYY, …)` would take the sample to roughly 30 race-cycles. That is a rate-limited API job against federal House races only, and it is the single highest-value data acquisition remaining in this series. Even then the sample stays small and uninstrumented — 30 observations with no exogenous variation still cannot identify a causal effect, only sharpen the descriptive picture. Genuinely settling the question needs a design this record cannot supply.

What it means

Money in Washington elections behaves like a **marker of candidate strength rather than a demonstrable mover of votes**. It correlates with winning more strongly than any other measured factor, and every attempt here to find the fingerprint a causal effect should leave comes up empty: allocation predicts nothing out of sample, the forecast model discards the term when the baseline is available, and the directional cross-section points the wrong way.

Two things follow, and they pull in opposite directions.

The first is a caution against the strong reform claim. "Money buys elections" is not supported by this evidence, and the donor paper's findings should not be read as implying it. What the donor paper establishes is that a narrow, old, geographically concentrated population *funds* campaigns — which matters for whose calls get returned and whose concerns reach an agenda, whether or not it moves a single vote. The companion cut in `cross-state-fec-money.md` §J sharpens this: in Washington's Solid seats the longshot party receives just **4.8%** of House inflow. Money overwhelmingly reaches candidates who were going to win anyway. That is a finding about *access*, not persuasion.

The second is a caution against the strong null. This paper has not shown money does not matter. It has shown that Washington's public record cannot demonstrate that it does, on the vote-moving axis specifically, at the sample sizes available. The **access and agenda-setting channel is entirely untested here** — it is the channel the donor paper's findings actually bear on, and the one most likely to matter. An honest reader should leave with the question open and a clear sense of what it would take to close it.

What this paper does not claim, and limits

- **This is not a causal estimate, and no causal claim is made in either direction.** Every cut is observational with no exogenous variation and no instrument. The +0.55 correlation is exactly what a real causal effect would also produce; the nulls are consistent with a true zero *and* with an effect the data is too thin to detect.
- **The allocation null is underpowered and tests the wrong thing twice over.** Coverage is 40 of 129 legislative baseline cells, holdout n runs 15–22, and it tests spend *mix*, not spend *level*. It cannot rule out an effect of simply having more money.
- **The IE cross-section is n = 7.** It is reported as description and should not be read as an estimate of anything. Its sign can change on one race.
- **State-legislative IE is excluded entirely** for want of a directional flag, so the largest available body of Washington IE never enters the analysis.
- **Only the persuasion channel is examined.** Access, agenda-setting, candidate entry deterrence, and the effect of money on *who runs in the first place* are all untested and all plausible routes by which money could matter without moving a general-election margin.
- **Coverage gaps favor larger campaigns.** PDC mini-reporting excludes the smallest campaigns, and 38% of expenditure dollars carry no purpose code.

Appendices

Appendix A — The objections, in full

1. "You found +0.55 and then explained it away." The correlation is real and the paper leads with it. The objection has force: the burden of proof for dismissing the largest correlation in the dataset should be high. What justifies not treating it as causal is not the correlation's size but the *pattern of everything around it* — allocation carrying no out-of-sample signal, the forecast model finding the term redundant against the baseline, and outside money associating with running behind. Any one of those alone would be weak. Together they describe money tracking a strength that already exists.

2. "Absence of evidence is not evidence of absence." Correct, and the paper's title question is answered "cannot tell," not "no." The allocation test is underpowered, the IE test is not estimable, and neither speaks to spend *level* with exogenous variation. This is stated in the body rather than buried, because the alternative — presenting a null as a finding of no effect — would be the more serious error.

3. "The negative IE slope is just strategic targeting, which you admit." Yes, and that is the point rather than a flaw. The negative association is offered as evidence of *endogeneity*, not of money being harmful. It demonstrates that the naive correlation cannot be read causally in either direction, which is the paper's central methodological claim.

4. "WA-03 is one race." It is, and it carries no inferential weight. It appears because it is the most extreme case available — the most IE-saturated House race in the country landing 0.06 points from its fundamentals — and because a reader who suspects the null is an artifact of small money should know what the largest observation looks like.

5. "Your baseline is model-derived, so you are testing your own model." Partly true and worth stating plainly. Overperformance and the IE residual are both measured against a fundamentals baseline this project built. That baseline is validated independently (`safe-seat-washington.md` shows its safe-seat projection matching observed results within a few points), but a reader who rejects the model should read Finding 1's correlation, which uses the neutral PVI baseline, and discount Findings 2b and 2c accordingly.

Appendix B — Data access and provenance

- **Election results.** Certified precinct-level returns from the Washington Secretary of State, aggregated to the seat. Public aggregate records.
- **Candidate finance.** Washington PDC candidate summaries and itemized C4 Schedule-A expenditures (Socrata `tijg-9zyp`), plus FEC candidate summaries for congressional races. Public disclosure records.
- **Independent expenditures.** FEC Schedule E for the 2024 cycle, carrying support/oppose and district; PDC independent-expenditure records for state legislative races, which carry amounts and races but **no directional flag**.
- **No personal data.** This paper uses candidate- and race-level aggregates only. It touches no voter file and releases no individual record.
- Full provenance and reproduction recipe: Data Sources & Reproducibility.

Appendix C — Methods

- **Overperformance.** `actual_dem_two_party_pct - baseline_dem_pct`, where the baseline is the model's neutral PVI plus down-ballot-drag figure, *not* the full prediction. Positive means the Democrat beat the district's fundamentals. Universe: the backtest grid of 10 congressional and 49 legislative districts across 2018–2024, restricted to the **163 baseline-scorable cells** — cells whose PVI fell back to a default for want of district presidential data are excluded, the same 163 used by the published backtest.
- **Fundraising feature.** Raw `log2(D receipts / R receipts)` matched by name tokens against `candidate_finance` for offices H/S/SR/SS, computed independently of the model's own capped fundraising term.
- **Allocation shares.** Field, media and professional shares of *coded operational* spend per candidate, so the measure is composition rather than level. Cross-cycle holdout fits 2022 and predicts 2024, the same bar used to reject the candidate-quality index earlier in this project.
- **IE residual test.** Net pro-Democratic IE = support-D plus oppose-R minus support-R minus oppose-D, per race, from FEC Schedule E. The dependent variable is the fundamentals-net residual, not the margin. Uncontested races and races without national-cycle data are dropped, leaving 7 of 17 candidate race-cycles scorable.
- **Inference threshold.** The IE script refuses to report a slope as inferential below 10 scorable races and prints its data-ceiling notice instead. That behavior is deliberate and should be preserved if the script is re-run after a backfill.
- **Reproduction.** `scripts/diag_overperformance_patterns.py` (Finding 1), `scripts/diag_expenditures_vs_residual.py` (2a), `scripts/diag_ie_vs_margin.py` (2c and Finding 3), `scripts/diag_loser_side_money.py` (the §J longshot-share figure).

Appendix D — Related work

This paper contributes a negative result and a data-availability audit, not a new identification strategy. It sits in a literature that has been circling the same endogeneity problem for five decades:

- **The endogeneity problem itself.** Jacobson, "The Effects of Campaign Spending in Congressional Elections," *American Political Science Review* 72(2) (1978): 469–491 — the founding observation that incumbent spending appears ineffective because incumbents raise and spend most when threatened. Jacobson, "The Effects of Campaign Spending in House Elections: New Evidence for Old Arguments," *American Journal of Political Science* 34(2) (1990): 334–362. The Washington pattern in Finding 2c is this problem reappearing in independent-expenditure data.
- **Designs that address it — and that this data cannot support.** Levitt, "Using Repeat Challengers to Estimate the Effect of Campaign Spending on Election Outcomes in the U.S. House," *Journal of Political Economy* 102(4) (1994): 777–798, doi:10.1086/261954 — differencing out candidate quality using repeated identical match-ups, and finding spending effects far smaller than naive estimates. Gerber, "Estimating the Effect of Campaign Spending on Senate Election Outcomes Using Instrumental Variables," *American Political Science Review* 92(2) (1998): 401–411 — the instrumental-variables approach. Both require panel structure or an instrument this record lacks.
- **The experimental literature.** Kalla & Broockman, "The Minimal Persuasive Effects of Campaign Contact in General Elections: Evidence from 49 Field Experiments," *American Political Science Review* 112(1) (2018): 148–166, doi:10.1017/S0003055417000363 — the best available evidence that general-election persuasion effects are

approximately zero on average, which is the prior this paper's nulls are consistent with.

- **Money as information rather than purchase.** Bonica, "Mapping the Ideological Marketplace," *American Journal of Political Science* 58(2) (2014): 367–386, doi:10.1111/ajps.12062 — contributions as a revealed-preference signal about candidates, the reading most consistent with the results here.
- **Access rather than votes.** Kalla & Broockman, "Campaign Contributions Facilitate Access to Congressional Officials: A Randomized Field Experiment," *American Journal of Political Science* 60(3) (2016): 545–558, doi:10.1111/ajps.12180 — the channel this paper explicitly does *not* test, and the one where an experimental effect has actually been demonstrated. Any reader inclined to conclude "money doesn't matter" from this paper should read that one.

Appendix E — The full IE cross-section

Every Washington congressional race with FEC Schedule-E data on disk, 2024 cycle (`scripts/diag_ie_vs_margin.py`). "Residual" is actual minus model-predicted Democratic share; a blank means the race was not scorable.

race	net pro-D IE	total IE	residual (pp)	actual margin
cd01 / 24	$0.00M	$0.00M	+7.59	+26.35
cd02 / 24	$0.00M	$0.00M	— *(uncontested)*	—
cd03 / 24	**+$16.18M**	**$40.08M**	**+0.06**	**+3.92**
cd04 / 24	+$1.88M	$6.77M	— *(uncontested)*	—
cd05 / 24	−$0.20M	$0.46M	−0.45	−21.25
cd06 / 24	+$5.69M	$5.69M	−3.52	+13.67
cd07 / 24	$0.00M	$0.00M	+13.70	+68.35
cd08 / 24	+$0.75M	$0.75M	+6.82	+8.18
cd09 / 24	$0.00M	$0.00M	— *(uncontested)*	—
cd10 / 24	$0.00M	$0.00M	−1.17	+17.35

Seven scorable races. Party attribution is complete — $0 of the $53.94M total is unresolvable to a side, so the negative slope is not a coding artifact. Note how much of the variation sits in races with **no** independent expenditure at all: cd07's +13.70 residual and cd01's +7.59 both occur at $0 IE, which is a compact illustration of why seven observations cannot separate money's effect from everything else that varies across districts.

End note — data, reproduction, and series

```
# Finding 1 — money vs overperformance across 163 race-cycles:
python scripts/diag_overperformance_patterns.py

# Finding 2a — spend allocation vs residual, cross-cycle holdout:
python scripts/diag_expenditures_vs_residual.py

# Findings 2c and 3 — IE vs fundamentals-net residual, and the data ceiling:
python scripts/diag_ie_vs_margin.py

# The longshot-share figure quoted in "What it means":
python scripts/diag_loser_side_money.py
```

All inputs are public records: certified election returns, WA PDC disclosure, and FEC bulk and API data. See `data-sources-and-reproducibility.md` for the full source ledger.

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